

One Poisson Equation for Conformal Coverage under Dependence: Exact State Correction and the Honesty Margin

Peter Cotton

`peter.cotton@microprediction.com`

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Abstract

Conformal prediction under dependent calibration data has two coupled defects. Dependence deforms the law of the order-statistic threshold, and the current dynamical state shifts the law of the future score. We show one Poisson equation controls both. The realized history-conditional coverage of the empirical-quantile threshold decomposes exactly into its invariant-law value plus a current-state corrector, and the same corrector supplies the martingale representation of the calibration error and its long-run variance. A Freedman bound yields a finite-sample, class-uniform honest quantile correction, and a union bound over a finite state space makes that correction state-adaptive at a $\log M$ cost, so the margin is paid at the state actually occupied rather than the worst one. In the fast-mixing limit the discrete corrector converges to a homogenization cell problem.

1 The exact object

Take a stationary time series of nonconformity scores and the split-conformal threshold, an order statistic of a calibration block. Two things are true at once and are usually conflated. The calibration scores are dependent, so the order statistic used as the threshold is not the one an i.i.d. analysis assumes. And the test point is generated by a dynamical process whose conditional law given the current state need not equal the invariant mixture. The exact object of study is neither the invariant law nor the marginal coverage. It is the realized history-conditional coverage of a random order-statistic threshold, and both effects act on it.

It is tempting to say that ordinary conformal prediction “calibrates the invariant law.” Under dependence that is not an exact guarantee. Dependence alters the law of the order statistic, and the conditional law given the current state alters the law of the future test score. Our contribution is that one solution family of a Poisson equation governs both perturbations: the transient gap between the current-state and invariant score laws, and the long-run calibration noise of the dependent empirical quantile. The identity is exact and asymptotics-free, and we verify every algebraic claim numerically to machine precision.

Contribution and related work. Two effects of this kind have been described before, and we treat them as background. Transported-beta analyses separate a test-side coverage transport from a deformation of the calibration order statistic (Ramos et al., 2026). Distributional conformal prediction for Markov processes constructs a conditional probability integral transform from estimated transition CDFs (Dai et al., 2026), and concentration-based penalties quantify coverage

loss under non-exchangeable and Markovian data (Oliveira et al., 2024; Zheng and Proutiere, 2024). Our narrow claim is this: we give a Poisson representation of realized history-conditional coverage and use the same solution family to obtain both the transient state correction and a finite-sample martingale-based tolerance correction for dependent calibration data. The nonparametric price of a fixed-point guarantee is the subject of a companion note (Cotton, 2026), whose local modulus reappears in Section 10.

2 The exact coverage decomposition

Let (W_t) be a Markov chain with transition kernel P and invariant distribution π , on a state space rich enough that all emission randomness is absorbed, so the score is a deterministic readout $R_t = r(W_t)$. This convention matters below: with the emission randomness inside W_t , the calibration-variance calculation needs no separate emission martingale. For each threshold q write the coverage function and the invariant score CDF

$$H_q(w) = \mathbf{1}\{r(w) \leq q\}, \quad F(q) = \pi H_q. \quad (1)$$

Assumption 1 (Ergodic chain with bounded Poisson solutions). The chain is uniformly ergodic with invariant law π , and there is a jointly measurable map $(w, q) \mapsto \chi_q(w)$ solving the Poisson equation

$$(I - P)\chi_q = H_q - F(q), \quad \pi \chi_q = 0, \quad (2)$$

with $\sup_q \|\chi_q\|_\infty < \infty$ over the quantile range used. For a finite irreducible aperiodic chain both hold automatically, and the reader may keep that case in mind.

Let $R_{(k)}$ be the k th order statistic of the calibration scores $R_1, \dots, R_n$, let $\mathcal{F}_n$ be the calibration history, and let $\mu_n(\cdot) = \mathbb{P}(W_n \in \cdot | \mathcal{F}_n)$ be the filtering distribution of the current state. If the test score is observed ℓ steps after the calibration block, its realized coverage is

$$C_{n,\ell,k} = \mathbb{P}\{R_{n+\ell} \leq R_{(k)} | \mathcal{F}_n\} = \mu_n(P^\ell H_{R_{(k)}}), \quad (3)$$

the ℓ -step transported coverage function averaged against the filter. The invariant law is the wrong reference by exactly the solved Poisson equation. Since $F(q)$ is constant and P^ℓ fixes constants,

$$P^\ell H_q - F(q) = P^\ell (H_q - F(q)) = (P^\ell - P^{\ell+1})\chi_q. \quad (4)$$

The first equality already writes the state discrepancy without solving anything. What the Poisson solution χ_q adds, and the reason we carry it, is that the *same* χ_q generates the additive-functional martingale and its variance in Section 5.

Theorem 1 (Exact conformal coverage corrector). *Under Assumption 1, for every calibration threshold $R_{(k)}$,*

$$C_{n,\ell,k} = F(R_{(k)}) + \mu_n[(P^\ell - P^{\ell+1})\chi_{R_{(k)}}]. \quad (5)$$

The first term is the coverage of the random calibration threshold under the invariant score law. The second is the exact current-state correction. If the state is observed at time n , so W_n is $\mathcal{F}_n$ -measurable, the filter is a point mass and

$$C_{n,\ell,k} = F(R_{(k)}) + (P^\ell - P^{\ell+1})\chi_{R_{(k)}}(W_n). \quad (6)$$

No asymptotics enter. Given the Poisson solution, (5) is an algebraic identity holding for every n, ℓ, k . A direct simulation on a finite chain confirms both sides of (6) agree to machine precision (Section 13).

3 A transported order-statistic representation

Suppose F is continuous and set $U_i = F(R_i)$. Each U_i is marginally uniform under stationarity, though the sequence need not be independent, and monotonicity of F gives $F(R_{(k)}) = U_{(k)}$. Define the state-dependent coverage transport

$$\tau_{\ell,w}(u) = P^\ell H_{F^{-1}(u)}(w) = u + (P^\ell - P^{\ell+1})\chi_{F^{-1}(u)}(w). \quad (7)$$

Then (6) reads $C_{n,\ell,k} = \tau_{\ell,W_n}(U_{(k)})$, which separates the problem cleanly. The law of $U_{(k)}$ carries the calibration dependence. The map τ_{ℓ,W_n} carries the test-side state dependence. Under i.i.d. calibration $U_{(k)}$ has the familiar beta law and τ is the identity, recovering ordinary conformal coverage. Under dependent calibration the law of $U_{(k)}$ is generally not beta, and its leading variance is set by the long-run variance of the coverage indicator, computed in Section 5. This is the transported-beta separation (Ramos et al., 2026), to which the Poisson representation adds the explicit corrector and a martingale representation of the calibration term.

4 The first-order coverage expansion

Fix $p = 1 - \alpha$ and $q_p = F^{-1}(p)$, and abbreviate the transient corrector

$$b_{\ell,w}(q) = P^\ell H_q(w) - F(q) = (P^\ell - P^{\ell+1})\chi_q(w). \quad (8)$$

The dependent empirical quantile obeys a Bahadur representation

$$F(R_{(k_p)}) - p = -[F_n(q_p) - p] + o_p(n^{-1/2}), \quad F_n(q) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{R_i \leq q\}, \quad (9)$$

valid under geometric ergodicity with F differentiable at q_p and $f(q_p) > 0$ (Glynn and Henderson, 2002). The expansion below differentiates the corrector in the threshold, so it needs more than (9) supplies. We assume in addition that, in a neighbourhood of q_p , the map $q \mapsto b_{\ell,w}(q)$ is continuously differentiable uniformly in w , that F is continuously differentiable with f bounded away from zero there, and that $R_{(k_p)} \rightarrow q_p$ in probability, which the same geometric ergodicity gives. For a finite chain the first condition reduces to continuous differentiability of each F_w near q_p , since $b_{\ell,w}$ is then a finite linear combination of the F_w . Under these, expanding (6) around q_p and using (9),

$$C_{n,\ell,k_p} - p = b_{\ell,W_n}(q_p) - \left[1 + \frac{1}{f(q_p)} \partial_q b_{\ell,W_n}(q_p)\right] [F_n(q_p) - p] + o_p(n^{-1/2}). \quad (10)$$

When the transient correction is small in C^1 , $\|b_{\ell,\cdot}\|_{C^1} = o(1)$, this reduces to the central formula

$$C_{n,\ell,k_p} - p = b_{\ell,W_n}(q_p) - [F_n(q_p) - p] + o_p(\|b_{\ell,\cdot}\| + n^{-1/2}). \quad (11)$$

Realized coverage error has two first-order sources, a current-state bias $b_{\ell,W_n}(q_p)$ and a calibration-CDF error $F_n(q_p) - p$. The first-order probability-level correction is $p \mapsto p - b_{\ell,W_n}(q_p)$. If the current state causes undercoverage then $b_{\ell,W_n}(q_p) < 0$ and the required empirical quantile level rises. A one-sided honesty margin is then added on top, and its size is set by the same operator.

5 One Poisson equation

No second Poisson equation is required. Center the coverage indicator at the invariant quantile, $g_p(w) = H_{q_p}(w) - p$. Because $F(q_p) = p$, equation (2) at $q = q_p$ reads $(I - P)\chi_{q_p} = H_{q_p} - p = g_p$, so by uniqueness of the centered solution the calibration-variance potential is the corrector itself,

$$\psi_p = \chi_{q_p}. \quad (12)$$

The single solution family $q \mapsto \chi_q$ therefore supplies both objects,

$$b_{\ell,w}(q) = (P^\ell - P^{\ell+1})\chi_q(w), \quad \sigma^2(q) = \pi[\chi_q^2 - (P\chi_q)^2]. \quad (13)$$

The additive calibration error has the exact martingale decomposition, indexed to match the calibration block $i = 1, \dots, n$,

$$\sum_{i=1}^n g_p(W_i) = P\chi_{q_p}(W_0) - P\chi_{q_p}(W_n) + \sum_{i=1}^n D_i, \quad D_i = \chi_{q_p}(W_i) - P\chi_{q_p}(W_{i-1}), \quad (14)$$

where (D_i) are martingale differences for the natural filtration and the endpoint term is bounded by $2\|\chi_{q_p}\|_\infty$. The long-run variance is

$$\sigma_p^2 = \sigma^2(q_p) = p(1-p) + 2 \sum_{j \geq 1} \text{Cov}_\pi(\mathbf{1}\{R_0 \leq q_p\}, \mathbf{1}\{R_j \leq q_p\}). \quad (15)$$

So one operator supplies the transient bias $b_{\ell,w}$ and the calibration variance σ_p^2 . These are not unrelated corrections. The corrector measures how much fast-state information survives to the test horizon. The variance measures how much fast-state persistence removes calibration information. For a reversible chain they are the same coefficients read two ways.

Proposition 2 (Reversible spectral reading). *Let P be reversible with respect to π and suppose its spectrum is pure point, with real eigenvalues $1 = \lambda_1 > \lambda_2 \geq \dots$ and a π -orthonormal eigenbasis ϕ_j of $L_0^2(\pi)$. A finite irreducible aperiodic chain, or any P with compact resolvent, satisfies this. For a chain with continuous spectrum, replace the sums below by integrals against the spectral measure of g_q , which leaves both identities intact in that form. Expand the centered coverage function $g_q = H_q - F(q) = \sum_{j \geq 2} c_j(q)\phi_j$. Then*

$$b_{\ell,w}(q) = \sum_{j \geq 2} c_j(q) \lambda_j^\ell \phi_j(w), \quad \sigma^2(q) = \sum_{j \geq 2} c_j(q)^2 \frac{1 + \lambda_j}{1 - \lambda_j}. \quad (16)$$

Proof. Solving (2) gives $\chi_q = \sum_{j \geq 2} c_j(q)(1 - \lambda_j)^{-1}\phi_j$, so $(P^\ell - P^{\ell+1})\chi_q = \sum_j c_j(q)\lambda_j^\ell(1 - \lambda_j)(1 - \lambda_j)^{-1}\phi_j$, the first identity. For the second, $\pi[\chi_q^2 - (P\chi_q)^2] = \sum_j c_j(q)^2(1 - \lambda_j^2)(1 - \lambda_j)^{-2} = \sum_j c_j(q)^2(1 + \lambda_j)/(1 - \lambda_j)$. $\square$

The same coefficients $c_j(q)$ and eigenvalues λ_j control the decay of current-state memory through λ_j^ℓ and the inflation of calibration variance through $(1 + \lambda_j)/(1 - \lambda_j)$. The slogan that state bias and effective-sample-size loss are two readings of one fast mode is exact when a single non-unit eigenvalue dominates, as in the two-state model below, and otherwise they are two functionals of the same spectrum.

6 A fast two-state volatility model

The cleanest one-mode instance is a fast volatility regime. Let $V_t^\varepsilon \in \{L, H\}$ be a continuous-time chain with generator

$$Q_\varepsilon = \frac{1}{\varepsilon} \begin{pmatrix} -a & a \\ b & -b \end{pmatrix}, \quad \kappa = a + b, \quad \pi_H = \frac{a}{a+b}, \quad \pi_L = \frac{b}{a+b}. \quad (17)$$

Conditional on the regime the score CDFs are $F_L(q)$ and $F_H(q)$, and the invariant score law is the mixture $F(q) = \pi_L F_L(q) + \pi_H F_H(q)$. We assume emissions are conditionally independent across time given the regime sequence, or equivalently that any remaining dependence has been folded into the augmented Markov state, so that (21) below holds. If the filtered high-state probability at time zero is $\omega_0 = \mathbb{P}(V_0 = H | \mathcal{F}_0)$, then at horizon t

$$\omega_t = \pi_H + e^{-\kappa t/\varepsilon}(\omega_0 - \pi_H), \quad (18)$$

so the exact conditional score CDF and transient coverage error are

$$G_t^\varepsilon(q) = F(q) + e^{-\kappa t/\varepsilon}(\omega_0 - \pi_H)[F_H(q) - F_L(q)], \quad b_t^\varepsilon(q) = G_t^\varepsilon(q) - F(q). \quad (19)$$

This is a literal homogenization boundary layer. After several fast mixing times it is exponentially small. At a forecast horizon comparable to ε it is not small at all. Where F is differentiable with $f > 0$ near q_p , and treating $e^{-\kappa t/\varepsilon}$ as small (that is, after enough fast mixing, not uniformly at $t = 0$), the quantile correction is

$$q_{t,p}^\varepsilon = q_p - \frac{b_t^\varepsilon(q_p)}{f(q_p)} + O(e^{-2\kappa t/\varepsilon}), \quad f = \pi_L f_L + \pi_H f_H. \quad (20)$$

For a nonnegative absolute-error score the high-volatility state usually has $F_H(q) < F_L(q)$ at the relevant q . If $\omega_0 > \pi_H$ then $b_t^\varepsilon(q_p) < 0$ and (20) raises the threshold, as it should.

Discrete sampling and calibration dependence. Sample every Δ time units. The nontrivial eigenvalue of the sampled transition matrix is $\lambda = e^{-\kappa\Delta/\varepsilon}$. Writing $d_p = F_H(q_p) - F_L(q_p)$, the coverage indicator $I_j = \mathbf{1}\{R_j \leq q_p\}$ has

$$\text{Cov}(I_0, I_j) = \pi_L \pi_H d_p^2 \lambda^j, \quad j \geq 1, \quad \sigma_p^2 = p(1-p) + 2\pi_L \pi_H d_p^2 \frac{\lambda}{1-\lambda}. \quad (21)$$

One fast eigenmode produces both effects, the transient bias $\propto \lambda^m(\omega_0 - \pi_H)d_p$ and the variance inflation $\propto \pi_L \pi_H d_p^2 \lambda/(1-\lambda)$, in agreement with Proposition 2 at a single mode. When $\Delta \ll \varepsilon$ we have $1 - \lambda \sim \kappa\Delta/\varepsilon$, so $\lambda/(1-\lambda) \sim \varepsilon/(\kappa\Delta)$. Sampling faster piles up redundant calibration scores, and for a window of fixed physical duration $T = n\Delta$ the effective sample size saturates at order T/ε rather than growing like n .

A numerical illustration. Take $p = 0.9$, $\pi_H = 0.2$, $\lambda = 0.8$, and at the invariant 0.9-quantile $F_L(q_p) = 0.95$, $F_H(q_p) = 0.70$, so the mixture is $0.8(0.95) + 0.2(0.70) = 0.90$. If the current state is high volatility the one-step coverage of the invariant threshold is $0.9 + 0.8(1 - 0.2)(0.70 - 0.95) = 0.74$. If it is low volatility it is $0.9 + 0.8(0 - 0.2)(0.70 - 0.95) = 0.94$. The same pooled threshold gives either 74% or 94% conditional coverage while appearing perfectly calibrated at 90% in stationarity. The long-run variance is $\sigma_p^2 = 0.09 + 2(0.8)(0.2)(0.25)^2(0.8/0.2) = 0.17$, an integrated

dependence factor $\sigma_p^2/[p(1-p)] \approx 1.89$, so the nominal calibration sample behaves like only $\approx 53\%$ as many independent observations. A direct simulation of the sampled chain reproduces σ_p^2 (Section 13). The persistent regime creates severe current-state undercoverage and nearly halves the effective calibration sample at once.

7 Homogenization: the continuous-time limit

The two-state calculation is the visible corner of a general fast–slow limit. Let Q be the generator of an ergodic continuous-time chain with invariant law π , sampled every Δ at fast scale ε , so the sampled kernel is

$$P_{\Delta,\varepsilon} = e^{(\Delta/\varepsilon)Q}, \quad \delta = \Delta/\varepsilon. \quad (22)$$

For the centered coverage function $g_q = H_q - F(q)$ define the continuous-time cell problem

$$-Q \varphi_q = g_q, \quad \pi \varphi_q = 0. \quad (23)$$

Proposition 3 (Poisson-to-cell convergence). *On the centered subspace, as $\delta \downarrow 0$,*

$$\delta \chi_q \longrightarrow (-Q)^{-1} g_q = \varphi_q, \quad (24)$$

where χ_q solves (2) for $P_{\Delta,\varepsilon}$. If Q is reversible, the per-observation long-run variance obeys $\delta \sigma_\delta^2(q) \rightarrow 2\langle g_q, (-Q)^{-1} g_q \rangle_\pi$, so for a calibration window of physical duration $T = n\Delta$,

$$\text{Var}(F_n(q)) \approx \frac{2\varepsilon}{T} \langle g_q, (-Q)^{-1} g_q \rangle_\pi. \quad (25)$$

At physical test horizon $t = m\Delta$ the transient corrector is the fast boundary layer $b_t^\varepsilon(q, w) = e^{tQ/\varepsilon} g_q(w)$.

Proof. On the centered subspace $I - e^{\delta Q} = -\delta Q(I + O(\delta))$, so $\delta(I - e^{\delta Q})^{-1} \rightarrow (-Q)^{-1}$ and $\delta \chi_q = \delta(I - e^{\delta Q})^{-1} g_q \rightarrow (-Q)^{-1} g_q$. For reversible Q , $\sigma_\delta^2(q) = \langle g_q, g_q \rangle_\pi + 2 \sum_{j \geq 1} \langle g_q, e^{j\delta Q} g_q \rangle_\pi$, a Riemann sum for $\delta^{-1} \int_0^\infty 2\langle g_q, e^{sQ} g_q \rangle_\pi ds = \delta^{-1} \cdot 2\langle g_q, (-Q)^{-1} g_q \rangle_\pi$. Dividing by $n = T/\Delta$ gives (25). The boundary layer is $P_{\Delta,\varepsilon}^m g_q = e^{m\Delta Q/\varepsilon} g_q$. $\square$

Three readings follow. The discrete Poisson solution converges to the cell solution. The honesty variance converges to the Green–Kubo coefficient. The current-state correction is the fast boundary layer. The two-state model of Section 6 is the one-mode corollary, and (25) is the general form of the effective-sample-size saturation at order T/ε . A slow state X_t^ε can be added through a frozen- x generator Q_x and its cell problem, with a remainder controlling the drift of x over the calibration and test horizons. We do not carry that step here.

8 A finite-sample honest Markov correction

The expansion motivates a finite-sample theorem. We ask for protection at one invariant probability level, not uniformly over every threshold, which is both sharper and cleaner.

Lemma 4 (One-sided Freedman corollary). *Let $(D_i)_{i \leq n}$ be martingale differences with $|D_i| \leq c$ and predictable quadratic variation $\sum_i \mathbb{E}[D_i^2 | \mathcal{F}_{i-1}] \leq nV$. Then for $x = \log(1/\eta)$, with probability at least $1 - \eta$, $\sum_{i=1}^n D_i \leq \sqrt{2nVx} + \frac{c}{3}x$.*

Write $x = \log(1/\eta)$ and, with $B = \|\chi_q\|_\infty$ and V a uniform bound on the conditional increment variance (specified per model below),

$$\delta_{n,\eta} = \frac{2B + \sqrt{2nVx}}{n} + \frac{2Bx}{3n}. \quad (26)$$

For a model class Θ of dependent score dynamics define the target-specific worst-case level

$$u^* = \inf \left\{ u \in [p, 1) : \inf_{\theta \in \Theta} \inf_w P_\theta^\ell H_{\theta, F_\theta^{-1}(u)}(w) \geq p \right\}, \quad (27)$$

the smallest invariant level such that, whatever the model and current state, the ℓ -step coverage of the corresponding invariant quantile is at least p .

Theorem 5 (Finite-sample class-uniform honest correction). *Suppose each F_θ is continuous and strictly increasing on the relevant range, that $u^* + \delta_{n,\eta} < 1$, and that the increments $D_i = \chi_{\theta, F_\theta^{-1}(u^*)}(W_i) - P_\theta \chi_{\theta, F_\theta^{-1}(u^*)}(W_{i-1})$ obey $|D_i| \leq 2B$ and conditional variance at most V , uniformly over $\theta \in \Theta$. Set $k = \lfloor n(u^* + \delta_{n,\eta}) \rfloor + 1$, with $R_{(k)} = +\infty$ if $k > n$. Then*

$$\inf_{\theta \in \Theta} \mathbb{P}_\theta \left\{ \mathbb{P}_\theta(R_{n+\ell} \leq R_{(k)} | \mathcal{F}_n) \geq p \right\} \geq 1 - \eta. \quad (28)$$

The procedure uses only the rank k , not the model.

Proof. Fix θ and write $u = u^*$, $q_u = F_\theta^{-1}(u)$, which is finite because $u < 1$ and F_θ is continuous and strictly increasing, and $F_\theta(q_u) = u$ so $g_u = H_{q_u} - u$ is centered. By (14) and Lemma 4, with probability at least $1 - \eta$,

$$F_n(q_u) \leq u + \delta_{n,\eta}.$$

On this event $nF_n(q_u) \leq n(u + \delta_{n,\eta}) < k$, so fewer than k calibration scores lie at or below q_u , hence $R_{(k)} \geq q_u = F_\theta^{-1}(u^*)$. Since $q \mapsto P_\theta^\ell H_q(w)$ is nondecreasing,

$$\mathbb{P}_\theta(R_{n+\ell} \leq R_{(k)} | \mathcal{F}_n) = P_\theta^\ell H_{R_{(k)}}(W_n) \geq P_\theta^\ell H_{F_\theta^{-1}(u^*)}(W_n) \geq p,$$

the last step by the definition (27) of u^* . The bound is uniform in θ . $\square$

Corollary 6 (Uniform-in-threshold form). *With $a_\ell = \sup_{w,q} [F(q) - P^\ell H_q(w)]_+$ and $u = p + a_\ell < 1$, the choice $u^* \leq p + a_\ell$ recovers the simpler calibration level $p + a_\ell$, since $P^\ell H_{F^{-1}(p+a_\ell)}(w) \geq F(F^{-1}(p+a_\ell)) - a_\ell = p$.*

This is a genuine calibration-sample honest and state-conditionally valid guarantee, stronger than a statement about average marginal coverage, and uniform over the class. One caution. The procedure is honest because k depends only on the fixed class constants u^*, B, V . If instead P, F , and the Poisson solution are estimated from the same data, finite-sample validity is not automatic. It then requires deterministic uniform class bounds, a simultaneous confidence set for the dynamics, or a separate estimation block with a valid envelope.

8.1 The state-adaptive correction

Theorem 5 protects against the worst current state. That is wasteful whenever the chain is usually somewhere cheaper, which is the normal case: a volatility regime that is dangerous one day in five still forces the one-day-in-five threshold on the other four. The corrector $b_{\ell,w}$ is a function of the state, so the natural procedure reads it at the state we are in. The obstacle is that the calibration rank then becomes random, and the Freedman step in Theorem 5 was applied at a single deterministic level. A union bound over the state space repairs this at a cost of $\log M$ inside the deviation.

Fix the observed-state case, so W_n is $\mathcal{F}_n$ -measurable and the filter is a point mass, and let the state space be finite with M elements. For each state w define the state-specific protection level

$$u^*(w) = \inf \left\{ u \in [p, 1) : \inf_{\theta \in \Theta} P_\theta^\ell H_{\theta, F_\theta^{-1}(u)}(w) \geq p \right\}, \quad (29)$$

which is (27) with the infimum over states removed. Then $u^*(w) \leq u^*$ for every w , and $u^* = \max_w u^*(w)$.

Assumption 2 (Attainment). For each w the infimum in (29) is attained, that is $\inf_{\theta} P_\theta^\ell H_{\theta, F_\theta^{-1}(u^*(w))}(w) \geq p$.

This holds whenever $u \mapsto \inf_{\theta} P_\theta^\ell H_{\theta, F_\theta^{-1}(u)}(w)$ is right continuous, in particular when each F_θ is continuous and strictly increasing and Θ is finite. Without it, replace $u^*(w)$ by $u^*(w) + \varepsilon$ throughout for any $\varepsilon > 0$, and every statement below goes through with that substitution. The same remark applies to u^* in Theorem 5.

Theorem 7 (State-adaptive honest correction). Assume the hypotheses of Theorem 5 and Assumption 2, with W_n observed and $|\mathcal{W}| = M < \infty$. Write $\delta_{n,\eta/M}$ for (26) evaluated at failure probability η/M , suppose $\max_w u^*(w) + \delta_{n,\eta/M} < 1$, and set the data-dependent rank

$$k(W_n) = \lfloor n(u^*(W_n) + \delta_{n,\eta/M}) \rfloor + 1, \quad (30)$$

with $R_{(k)} = +\infty$ if $k > n$. Then

$$\inf_{\theta \in \Theta} \mathbb{P}_\theta \left\{ \mathbb{P}_\theta(R_{n+\ell} \leq R_{(k(W_n))} \mid \mathcal{F}_n) \geq p \right\} \geq 1 - \eta, \quad (31)$$

and $k(W_n) \leq k$ for every realization, with strict inequality whenever $u^*(W_n) < u^* - (\delta_{n,\eta/M} - \delta_{n,\eta})$.

Proof. Fix $\theta \in \Theta$. For each state w the level $u^*(w)$ is a deterministic constant of the class, so $q_w := F_\theta^{-1}(u^*(w))$ is deterministic and $g_w = H_{q_w} - u^*(w)$ is centered under π , because $F_\theta(q_w) = u^*(w)$. Applying the martingale decomposition (14) and Lemma 4 at level $u^*(w)$ with failure probability η/M gives, for that single w ,

$$\mathbb{P}_\theta \{ F_n(q_w) > u^*(w) + \delta_{n,\eta/M} \} \leq \eta/M.$$

Union over the M states: on an event E of probability at least $1 - \eta$,

$$F_n(F_\theta^{-1}(u^*(w))) \leq u^*(w) + \delta_{n,\eta/M} \quad \text{simultaneously for every } w \in \mathcal{W}. \quad (32)$$

The event E does not depend on which state is realized, which is what makes the random rank admissible.

Work on E and let $w = W_n$ be the realized state. By (32) at that w ,

$$nF_n(q_w) \leq n(u^*(w) + \delta_{n,\eta/M}) < k(w),$$

so strictly fewer than $k(w)$ calibration scores lie at or below q_w , and therefore $R_{(k(w))} \geq q_w = F_\theta^{-1}(u^*(w))$. Since $q \mapsto P_\theta^\ell H_q(w)$ is nondecreasing, and using (6) for the realized coverage,

$$\mathbb{P}_\theta(R_{n+\ell} \leq R_{(k(w))} \mid \mathcal{F}_n) = P_\theta^\ell H_{R_{(k(w))}}(w) \geq P_\theta^\ell H_{F_\theta^{-1}(u^*(w))}(w) \geq p,$$

the last inequality by Assumption 2. The chain of inequalities used no property of θ beyond the class hypotheses, so the bound is uniform over Θ . The comparison $k(W_n) \leq k$ is immediate from $u^*(W_n) \leq u^*$ once $\delta_{n,\eta/M} - \delta_{n,\eta} \leq u^* - u^*(W_n)$. $\square$

Two things are worth separating. The procedure is state-adaptive but not model-adaptive: the map $w \mapsto u^*(w)$ is a fixed class constant computed before seeing data, exactly as u^* is in Theorem 5, and only its *argument* is read off the realized state. The caution at the end of Section 8 therefore still applies unchanged. If P , F and χ are estimated from the same calibration block, finite-sample validity is not automatic.

Corollary 8 (When adapting pays). *The adaptive rank is strictly smaller than the worst-case rank at state w whenever*

$$u^* - u^*(w) > \delta_{n,\eta/M} - \delta_{n,\eta} = \sqrt{\frac{2V}{n}} \left(\sqrt{\log \frac{M}{\eta}} - \sqrt{\log \frac{1}{\eta}} \right) + \frac{2B}{3n} \log M. \quad (33)$$

The right side is $O(\sqrt{V \log M/n})$ and the left side is a constant of the class, so the adaptive procedure is better at every state off the worst case once $n \gtrsim V \log M / (u^ - u^*(w))^2$, and the expected saving in calibration level tends to $\mathbb{E}_\pi[u^* - u^*(W)]$.*

The two-state model. Take the illustration of Section 6, with $p = 0.9$, $\pi_H = 0.2$, $\lambda = 0.8$, $\ell = 1$, and regime score laws chosen so that $F_L(q_p) = 0.95$ and $F_H(q_p) = 0.70$ at $q_p = 1$. Solving (29) gives

$$u^*(L) = 0.9000, \quad u^*(H) = 0.9724, \quad u^* = 0.9724.$$

The low state needs no correction at all, because the invariant threshold already over-covers there. In threshold units the worst-case procedure calibrates at $F^{-1}(0.9724) = 1.774$ while the adaptive procedure calibrates at $F^{-1}(0.9) = 1.000$ in the low state, a 44% narrower set, and the chain is in the low state 80% of the time. The expected saving in level is $\mathbb{E}_\pi[u^* - u^*(W)] = 0.058$. With the exact conditional increment variance the crossover in (33) is near $n = 55$, so the adaptive rank is the smaller one over the whole practical range. A direct simulation of the sampled chain confirms the guarantee (31): at $\eta = 0.10$ the realized failure rate of the adaptive procedure is 0.000 at $n \in \{500, 2000, 8000\}$ against the worst-case procedure's 0.000 to 0.002, both far inside the nominal level, as Freedman's inequality is conservative (Section 13).

The gap is not a refinement of constants. At $n = 1000$ the worst-case level $u^* + \delta_{n,\eta}$ exceeds one, so $R_{(k)} = +\infty$ and the worst-case set is vacuous, while the adaptive level in the low state is 0.94 and the set is finite. Adapting to the state is what makes the guarantee usable at that sample size.

Remark 9 (Latent state). When W_n is not observed, $\mathcal{F}_n$ determines only the filter $\mu_n(\cdot) = \mathbb{P}(W_n \in \cdot | \mathcal{F}_n)$, and (5) rather than (6) is the relevant identity. Theorem 7 does not apply as stated, because u^* would have to be indexed by a point of the simplex rather than by one of M states, and the union bound over a continuum is not available without a covering argument and a modulus for $\mu \mapsto u^*(\mu)$. Two honest options remain. Index by the support of the filter, using $\max\{u^*(w) : \mu_n(w) > 0\}$, which recovers Theorem 7 when the filter is degenerate and Theorem 5 when it is fully supported. Or partition the simplex into finitely many cells before seeing data, index u^* by cell, and pay $\log(\text{number of cells})$ in (33). We do not pursue the second here.

9 Interpretation

The corrected empirical quantile level is

$$p + \underbrace{a_\ell}_{\text{fast-state protection}} + \underbrace{\delta_{n,\eta}}_{\text{dependent-calibration protection}}, \quad (34)$$

and in threshold units the first-order inflation is $(a_\ell + \delta_{n,\eta})/f(q_p)$. The two costs are qualitatively different. The transient term a_ℓ is not an estimation error. It is the real difference between the current conditional law and the invariant law. The term $\delta_{n,\eta}$ is the price of demanding that the realized calibration sample be trustworthy with probability $1 - \eta$. A state-adaptive implementation need not pay the worst-case a_ℓ . It can estimate $b_{\ell, W_n}(q) = (P^\ell - P^{\ell+1})\chi_q(W_n)$ and build a lower confidence envelope for the current conditional CDF. Theorem 5 is the simplest version with a transparent proof.

10 Combining with the local sharpness modulus

The companion note (Cotton, 2026) shows that honest coverage at a fixed covariate value x_0 forces inflation at the local nonparametric modulus $(ng(x_0))^{-s/(2s+d)}$, from locally indistinguishable alternatives rather than any exchangeability phenomenon. The dependent extension is schematic, and we label it as such. For a kernel K_h the relevant local object is the weighted long-run variance

$$\Omega_h(x_0) = \sum_{j \in \mathbb{Z}} \text{Cov}\left(K_h(X_0 - x_0)[\mathbf{1}\{R_0 \leq q_p(x_0)\} - p], K_h(X_j - x_0)[\mathbf{1}\{R_j \leq q_p(x_0)\} - p]\right), \quad (35)$$

and the argument needs a local Markov mixing CLT of the form $h^d \Omega_h(x_0) \rightarrow \omega_p(x_0)$.

Remark 10 (Schematic bias–margin decomposition). Under a fixed uniform spectral gap the effective local sample stays of order $ng(x_0)h^d$, and the natural upper bound is, in threshold units,

$$T_{n,h}(x_0) - q_p(x_0) \lesssim C_q L h^s + \frac{\sqrt{\omega_p(x_0) \log(1/\eta)}}{f_{p,x_0} \sqrt{ng(x_0)h^d}} + \frac{a_{\ell,x_0}}{f_{p,x_0}} + \frac{R_{\text{hom}}(\varepsilon) + \Delta_{\text{dyn}}}{f_{p,x_0}}, \quad (36)$$

all terms on the threshold scale. Minimizing the first two over h gives $h_\star \asymp [\omega_p(x_0)/(ng(x_0)L^2)]^{1/(2s+d)}$ and local inflation $\asymp L^{d/(2s+d)}[\omega_p(x_0)/(ng(x_0))]^{s/(2s+d)}$.

So under a fixed spectral gap, dependence changes the information constant through the local long-run variance but leaves the Hölder exponent unchanged. If the mixing time itself diverges

with n , h , or ε , then $\omega_p(x_0)$ may diverge and the rate can change. Homogenization adds the transient and dependence terms and does not erase the local modulus. The hierarchy is

$$\begin{aligned}
&\text{local heterogeneity} \rightarrow \text{nonparametric modulus,} \\
&\text{calibration persistence} \rightarrow \text{Poisson long-run variance,} \\
&\text{current fast state} \rightarrow \text{Poisson transient corrector,} \\
&\text{approximating the fast-slow model} \rightarrow \text{homogenization remainder.}
\end{aligned} \tag{37}$$

The first term is unavoidable over a broad Hölder class. The second and third can be quantified and corrected under dynamical structure. The fourth must be included for the guarantee to be honest.

11 What homogenization does and does not buy

Homogenization does not defeat the sharpness lower bound. It partitions the conformal error into unavoidable local ambiguity, correctable dynamical bias, and estimable dependence noise. If the fast-slow dynamics live in a finite-dimensional parametric family, honest uncertainty about the dynamical correction shrinks at the parametric rate $n^{-1/2}$ rather than the nonparametric local rate $n^{-s/(2s+d)}$. That gain does not contradict the companion note. It comes from replacing a broad Hölder class by a much smaller structured experiment. If the transition law, the state-specific score distributions, or the scale field are again allowed to vary nonparametrically with x , the local modulus returns, and if their smoothness is unknown the no-adaptation obstruction returns with it.

12 The corresponding skaters transform

The correction is directly usable as a recalibration layer. For an existing distributional forecast F_{t+1}^0 , compute the Gaussianized probability integral transform $Z_t = \Phi^{-1}(F_t^0(Y_t))$, and fit a predictable two-state model to the past Z_t with state-specific CDFs K_L, K_H and filtered high-state probability ω_t . With $\omega_{t+1|t} = \pi_H + \lambda(\omega_t - \pi_H)$ the conditional CDF of the next Gaussianized PIT is $K_{t+1|t}(z) = (1 - \omega_{t+1|t})K_L(z) + \omega_{t+1|t}K_H(z)$, and the dynamically corrected predictive CDF and u -quantile are

$$F_{t+1}(y) = K_{t+1|t}[\Phi^{-1}(F_{t+1}^0(y))], \quad \hat{Q}_{t+1}(u) = (F_{t+1}^0)^{-1}[\Phi(K_{t+1|t}^{-1}(u))]. \tag{38}$$

This is a predictable dynamic PIT recalibration, not by itself a finite-sample conformal procedure. The fence-post map estimates a pooled PIT law, and (38) estimates the conditional PIT law after accounting for the current residual regime. To connect it to Theorem 5 the implementation needs a predictable or separately estimated regime model, an uncertainty envelope covering parameter and filtering error, a probability-level margin, and a treatment of ties or discrete forecast distributions. For an honest upper quantile, let $\mathcal{C}_t$ be a confidence set for the dynamic parameters and add a homogenization or misspecification radius r_t ,

$$z_{t+1,p}^{\text{hon}} = \sup_{\theta \in \mathcal{C}_t} K_{\theta,t+1|t}^{-1}(p + r_t), \quad y_{t+1,p}^{\text{hon}} = (F_{t+1}^0)^{-1}[\Phi(z_{t+1,p}^{\text{hon}})]. \tag{39}$$

Because distributional conformal prediction for Markov processes already composes a transition-CDF PIT (Dai et al., 2026), the novelty here is the Poisson state correction and its honesty envelope, not the conditional PIT composition itself.

Empirical protocol. A comparison should include the base forecast, the marginal fence-post correction, an AR correction, the two-state homogenized correction, and the honest two-state correction. The decisive diagnostics are PIT calibration conditional on the filtered state, coverage conditional on recent Z_t and volatility, the lower-tail probability of realized coverage, proper scores such as log score and CRPS, and the extra interval width the honest envelope demands. Aggregate PIT uniformity alone is insufficient, since the two-state example shows pooled 90% calibration concealing conditional coverage from 74% to 94%.

13 Reproducibility

Two short scripts accompany the note. `verify_decomposition.py` checks, on random finite chains, the corrector identity (6) (maximum error 5.6×10^{-16}), the identity $\psi_p = \chi_{q_p}$ of (12) (exactly zero), the reversible spectral identities of Proposition 2 (error $\sim 10^{-16}$), and the generator limit of Proposition 3 (error decaying linearly in δ). `two_state_illustration.py` reproduces the 74%/94% conditional coverages, the long-run variance 0.17, and the effective sample fraction 0.53, and confirms σ_p^2 by direct simulation of the sampled chain. `state_adaptive.py` recomputes the illustration from the regime score laws, solves (29) for $u^*(L)$, $u^*(H)$ and u^* , checks the guarantee (31) by simulating the sampled chain at $n \in \{500, 2000, 8000\}$ with $\eta = 0.10$, and tabulates the crossover in (33).

14 Conclusion

The conformal coverage process under Markov dependence admits an exact invariant-plus-Poisson decomposition, and one solution family $q \mapsto \chi_q$ determines both the current-state coverage bias and the dependence-adjusted honesty margin. State bias and effective-sample-size loss are two readings of the same spectrum, one mode in the two-state case. Because the corrector is a function of the state, the honest margin can be read at the state actually occupied rather than at the worst one, and Theorem 7 shows a union bound over a finite state space buys that at a log M cost inside the deviation. In the two-state illustration the adaptive threshold is 44% narrower than the worst-case threshold in the state the chain occupies four fifths of the time, and it is finite at sample sizes where the worst-case set is vacuous. In the fast-mixing limit the discrete Poisson solution becomes a homogenization cell problem, the calibration variance becomes a Green–Kubo coefficient, and the state correction becomes the fast boundary layer. Placed next to the local sharpness modulus, this identifies what dynamical structure can and cannot buy back. It corrects the transient and dependence terms, it does not remove the nonparametric floor, and honest use requires paying for the approximation of the fast–slow model itself.

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